

Digits

Tags:

Time Limit: 1 s Memory Limit: 128 MB
Submission: 65 AC: 35 Score: 98.48

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Description

A googol written out in decimal has 101 digits. A googolplex has one plus a googol digits. That's a lot of digits!

Given any number x_0 , define a sequence using the following recurrence:

$$x_{i+1} = \text{the number of digits in the decimal representation of } x_i$$

Your task is to determine the smallest positive i such that $x_i = x_{i-1}$.

Input

Input consists of several lines. Each line contains a value of x_0 . Every value of x_0 is non-negative and has no more than one million digits. The last line of input contains the word `END`.

Output

For each value of x_0 given in the input, output one line containing the smallest positive i such that $x_i = x_{i-1}$.

Samples

input	Copy
42	
END	
output	Copy
3	

Submit

Codes

